

A Simple Bi-directional Bridgeless AC/DC Buck-Boost Converter for Automotive Energy Harvesting

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Abstract—In this paper, a new switched-mode bidirectional bridgeless AC/DC converter, operating in the discontinuous conduction mode (DCM), for automotive energy harvesting is presented. The simple topology consisting of common source n-MOSFETs allows for direct AC/DC conversion, minimum auxiliary circuits and simple gating patterns, when compared to other AC/DC converters proposed in the literatures. The circuit operation is analyzed and shown that the converter-battery topology can operate as a synthesized variable resistor with regenerative functionality. The circuit is designed for low-power energy harvesting applications; for example, regenerating dissipated power in a vehicle suspension damper by converting vibrational energy into battery charge. Simulation results including various switching waveforms and power conversion efficiency are presented to demonstrate performance of the proposed converter. Experimental results, including power conversion efficiency and resistance synthesis sweep, by connecting the proposed converter to a regenerative vehicle suspension prototype are also included.

Keywords— *Low- power energy harvesting; Bi- directional bridgeless AC/DC; Regenerative vehicle suspension;*

I. INTRODUCTION

Various types of energy harvesting systems have been reported in the literatures, including electromagnetic- based generators and AC/DC converters for vibrational energy harvesting [1]- [7]. In addition, a two-stage converter consisting of a buck converter with a full-bridge rectifier was proposed in [8]-[10] for low- power/ high line voltage (piezoelectric-based) vibrational energy harvesting. A control method for synthesizing variable input resistance using a bridgeless boost rectifier was proposed in [11]. In [12], another piezoelectric-based energy harvester was presented by utilizing a buck and buck-boost converter. Method of impedance matching for maximum power transfer in a piezoelectric-based energy transducer was presented in [13]. A fly-back converter with maximum power point tracking capability was presented in [14]. A MOSFET-based voltage doubler coupled with a boost converter was proposed in [15] for low voltage energy harvesting applications. In [16]-[17], the concept of energy harvesting by providing an equivalent damping was presented by utilizing a full-bridge rectifier and a DC/DC converter. A boost type converter designed for low-frequency vibrating energy harvesting for wireless sensor nodes was investigated in [18]. A low- voltage/ power direct AC/DC converter operating in DCM is proposed in [19]. It

utilized a bi- directional MOSFET and minimal auxiliary circuits for eliminating the placement of full- wave rectifier and enhancing the conversion efficiency.

While the above works have contributed to developing power converters for applications in low- power energy harvesting (i.e. < 5 W), certain drawbacks exist especially for two-stage AC/DC converter configurations [8- 10], [12], [20]. For instance, the forward voltage drops (≈ 0.5 V for Schottky diodes and ≈ 0.8 V for typical PN-junction diodes) in the line diodes result in conduction power losses and consequently efficiency degradation. To alleviate this problem, either an integrated bridge or synchronous rectification topology can be utilized. However, there are certain drawbacks in using these alternatives. For example, integrated bridge rectifiers generally are limited by continuous current conduction level. For example, HD04-T from Diodes Incorporated [20] operates below maximum continuous current of 0.8A. In addition, conventional synchronous and two- stage buck/ buck-boost topologies require high-side gate driver IC for driving high side n-MOSFETs and requiring more complex control method. As a result, the power consumption inevitably reduces the power conversion efficiency especially in low- power energy harvesting applications.

This paper presents a new bridgeless configuration, operating in DCM. The converter can function as a sensorless variable synthesized resistor for battery charging, which makes it practical when connected to irregular line voltage induced in a seismic environment; also, the converter is capable of powering DC load with fixed voltage rating through voltage mode control (VMC). Due to nature of sensorless based, the proposed converter is able to synthesize variable resistor even at low step- up ratio (between line and load voltage) in DCM operation comparing to [19], which leads to smaller (lower power consumption assuming same power level applications) auxiliary circuit foot- print.

The circuit is designed for applications in low- power energy harvesting; for example, it regenerates energy from a vibrational power source by providing variable electrical damping. Synthesizing electrical damping is similar to achieving unity power factor between line voltage and current (e.g., [21]- [23]) for maximum available power transfer. The proposed topology is a modified buck-boost AC/DC converter

without line diodes in a full-bridge configuration. Avoiding line diodes and having simple gating pattern would enable it to operate with lower standby power loss and simpler control method compared to conventional synchronous and two-stage AC/DC topologies [8]-[10], designed for similar power level operations. Moreover, due to its simple control method, the requirement of auxiliary components is minimized resulting in a more compact converter layout, making it ideal to integrate/retrofit with a vehicles suspension.

The organization of this paper is as follows. Section II provides the detailed topology and operation modes of the proposed converter. Section III presents the analysis and modeling of the proposed converter under various switching modes. Section IV presents simulation results of the converter for theoretical analysis verification along with the simulated power conversion. Brief experimental results obtained by coupling the converter to a regenerative vehicle suspension prototype is documented in section V. Conclusions are presented in section VI.

II. PROPOSED BRIDGELESS CONVERTER TOPOLOGY

The proposed converter topology, shown in Fig. 1(a), is a modified direct AC/DC buck-boost converter operating in DCM with bi-directional current conduction and blocking capability. The configuration is inspired by battery charging circuits, where back-to-back MOSFETs are utilized for over-charge and over-discharge protection [25]- [26]. This configuration is utilized to ensure line voltage v_{in} is disconnected to inductor L during its' discharging period in the entire line cycle. The back-to-back configuration is placed intentionally to prevent undesired conduction via the undesired path. For example, if SW_{MOS} only consisted of a single MOSFET with its' drain connected to v_{in} , during negative line cycle, even if SW_{MOS} is turned off, the body diode will still be forward biased if v_{in} is higher than body diode forward voltage V_F .

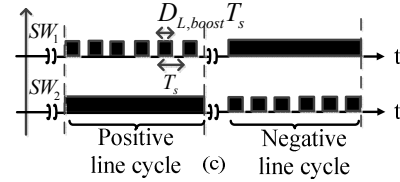
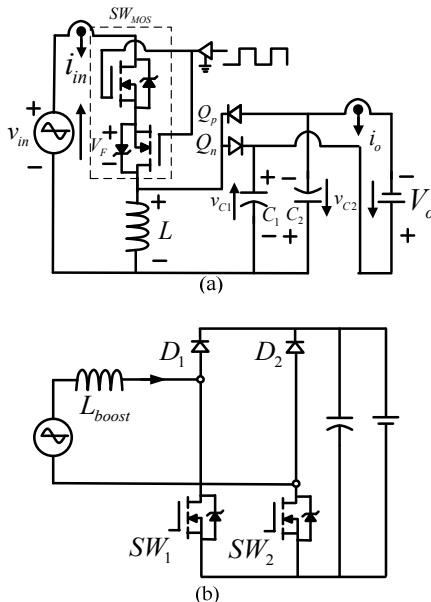


Fig. 1(a) Proposed bi- directional bridgeless AC/DC converter. (b) Conventional two- switch type synchronous rectifier with its' corresponding (c) gating signal for DCM operation.

To synthesize an adjustable variable input resistor, both the line voltage v_{in} and current i_{in} should be made in-phase through proper control. DCM operation results to resistive behavior between v_{in} and i_{in} and minimizing the turn- on switching losses of the MOSFETs with zero current switching (ZCS). In addition, due to longer recovery time, the requirements of soft factor and reverse recovery t_{rr} of rectifying diodes are relaxed.

Comparing to a synchronous boost converter [11], also operating in DCM and implemented for a low- power regenerative suspension prototype, the main advantages are as follows. In DCM, the conduction loss in power stage is the primary contributor of power efficiency degradation. The proposed converter avoids additional conduction loss contributed by $R_{DS,on}$ of either SW_1 or SW_2 in synchronous boost converter (see Fig. 1(b)) during the inductor discharging period. Moreover, the synthesized input resistance is a function of time- independent variables (shown later). In addition, gating pattern of the proposed topology is extremely simple, where both n- MOSFETs are gated simultaneously. On the contrary, either SW_1 or SW_2 is required to switch on for entire half line cycle, which leads to more complex gating pattern and line polarity detection (see Fig. 1(c)) is needed to realize such operation.

III. CONVERTER ANALYSIS AND MODELLING

In this section, the switching waveforms of the proposed converter are analyzed by assuming ideal MOSFET switching, zero forward voltage drop across the diodes, and zero series resistances for passive elements. The comparisons of the proposed power converter with a conventional boost rectifier in terms of various equivalent circuits under different operation modes are presented.

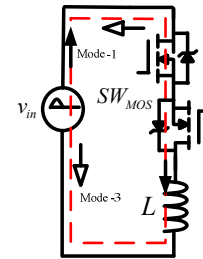


Fig. 2: Mode 1/3 equivalent circuit.

Referring to Fig. 1, it is assumed that all n-MOSFETs (SW_1 , SW_2 , and SW_{MOS}), and rectifying diodes (D_1 , D_2 , Q_p , and Q_n), are identical. The converter is designed to operate in various

conduction modes during both positive and negative cycles of voltage v_{in} . Conduction mode-1/ 3 and 2/ 4 represents charging and discharging of line inductor L during positive and negative line cycle, respectively. The details are outlined in the following.

A. Mode-1/ 3

The equivalent circuit with the current flow for Mode-1 is shown in Fig. 2. To illustrate the switching waveforms in different switching modes, line current i_{in} , average line current $i_{in,avg}$, average output current $i_{o,avg}$, diode current, inductor current i_L , inductor voltage v_L , line voltage v_{in} , battery voltage V_o , capacitor current, and duty cycle of SW_{MOS} are shown in Fig. 3. In addition, i_{pk} , $i_{in,avg}$, $I_{L,boost}$, $\Delta i_{L,boost}$ are the peak inductor current, average line current, average inductor current, inductor current ripple, respectively. The duty cycle represents inductor charging in both line cycles is D_L . The discharging duty cycle in both positive and negative line cycles are indicated by D_p and D_n , respectively. T_i and T_s are line cycle and switching period, respectively.

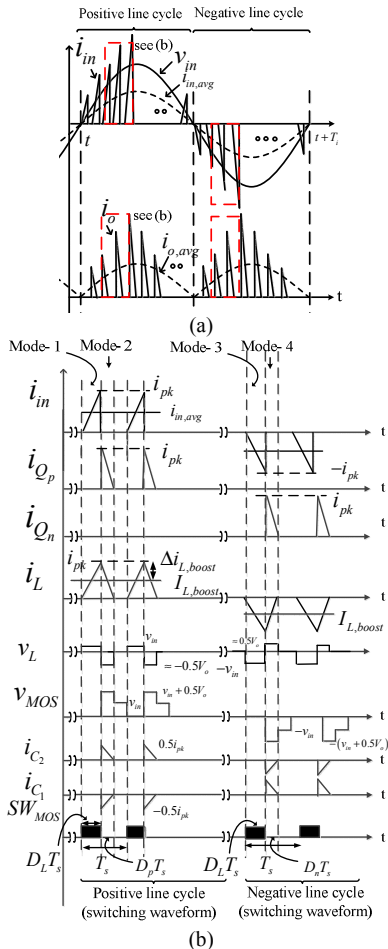


Fig. 3: (a) Ideal theoretical switching waveforms in single line cycle and (b) switching periods.

Note that other than opposite current flow, in order to facilitate simple gating pattern, the proposed topology is designed to operate identically in Modes 1/ 3 and 2/4. In Mode 1, SW_{MOS} is switched on, with the top n- MOSFET operating

in forward conduction and the bottom MOSFET in the reverse conduction mode. The charging of L will take place in Mode 1 since $V_{L,boost} = v_{in} > 0$. Since both Q_p and Q_n are reverse biased, current does not flow into the load.

B. Modes 2/ 4

In Mode- 2, n- MOSFETs SW_{MOS} are switched off to allow the energy stored in L to discharge into the load, as depicted in Fig. 4 (a). For KVL to hold, the total voltage across both C_1 and C_2 equals to V_o . As a result, as the current charges both C_2 and the battery, C_1 is discharged. In this mode, $V_L = v_{in} - V_o < 0$; hence, the rectifying diode Q_p is forward biased for the duration $D_p T_s$ in order to charge the load and ensure validity of volt-sec balance principle. After i_L reaches zero, $V_L = 0$ and the entire converter is essentially switched off. The current conduction of Mode- 4 for both converters, shown in Fig. 5, is similar to that of Mode- 2.

In contrast, the synchronous boost converter requires gate drive to remain logic high in order for SW_2 to form a closed-loop as shown in Fig. 4 (b). Therefore, in Mode- 2 the power dissipation/conduction losses are higher due to $R_{DS,ON}$ of SW_2 . The gating pattern also leads to unequalled duty cycles/ stress between both n- MOSFETs and potentially limits choosing the modulation carrier frequency and MOSFET's safe operating area (SOA).

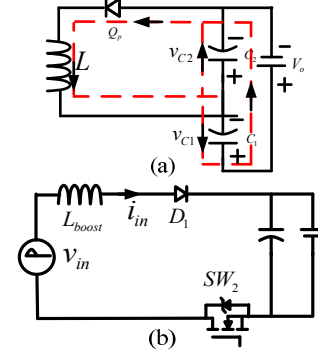


Fig. 4: Mode-2 equivalent circuits of (a) bridgeless AC/DC converter, (b) conventional boost rectifier during positive line cycle.

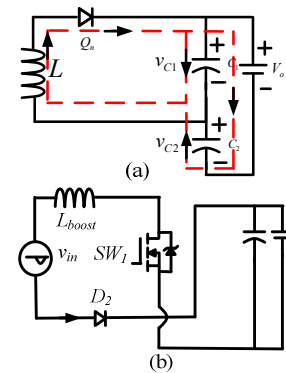


Fig. 5: Mode-4 equivalent circuits of (a) bridgeless AC/DC converter, (b) conventional boost rectifier during negative line cycle.

IV. CONVERTER ANALYSIS

As mentioned previously, synthesizing variable resistors is equivalent to providing mechanical damping in the electrical

domain. The equivalent variable input resistance can be defined in terms of line voltage to either envelope or average input current as envelope input resistance or average input resistance, respectively. Before deriving the expressions for input resistors synthesis seen from the line terminal the average input power of the proposed converter is obtained by **averaging instantaneous input power (i.e. E_{in}/T_s) over half line cycle (i.e. $T_i/2$)** as the following

$$P_{in} = \frac{2}{T_i} \int_0^{T_i/2} \frac{E_{in}}{T_s} dt = 0.25 D_{L,boost}^2 T_s V_p^2 L^{-1} \quad (1)$$

where V_{in} is line voltage amplitude, E_{in} is the converter input energy, T_i is line voltage period, and T_s is switching period. Similarly, average input power of the synchronous boost converter is derived as

$$P_{in,SB} = \int_0^\pi \left(\frac{D_{L,boost}^2 T_s}{2L_{boost}} \right) \frac{(V_{in} \sin \theta)^2}{1 - \frac{V_{in}}{V_o} \sin \theta} d\theta \approx 0.25 V_{in}^2 D_{L,boost}^2 T_s L^{-1} \quad (2)$$

where V_o is load voltage and $\theta = 2\pi T_i^{-1} t$. Note that the input power in (2) can only be approximated, since the instantaneous term, $A = \int_0^\pi \sin(\theta)^2 (1 - V_{in} V_o^{-1} \sin \theta)^{-1} d\theta \approx 0.5\pi$ when $V_{in} \ll V_o$ is assumed as shown in Fig. 6. Comparing (1) and (2), the input power of the proposed converter is solely a function of input voltage (assuming other parameters remain constant); thus making it synonymous to that of physical resistor and ideal for resistor synthesis. On the contrary, input power of the synchronous boost converter is a function of V_{in} and V_o , making the synthesized resistor nonlinear (shown later).

The average line current $i_{in,avg}$ and rectified current $i_{o,avg}$, shown in Fig. 3(a), are obtained as the following

$$i_{in,avg} = 0.5 v_{in} D_{L,boost}^2 T_s L^{-1} \quad (3)$$

$$i_{o,avg} = v_{in}^2 D_{L,boost}^2 T_s L^{-1} V_o^{-1}. \quad (4)$$

From (3) - (4), it is observed that $i_{in,avg}$ oscillates with respect to T_i while $i_{o,avg}$ is summation of both DC and second harmonics $(2T_i)^{-1}$ terms. In addition, to ensure DCM operation, maximum duty cycle of gating pulses is also derived as

$$D_{L,max} < V_p V_o (V_p V_o + 2V_p^2)^{-1}. \quad (5)$$

Combining (1) - (4), the average line resistance of the proposed converter (i.e., R_{in}) and conventional synchronous boost rectifier (i.e., $R_{in,SB}$) are defined as follows

$$R_{in} = \frac{v_{in}}{i_{in,avg}} = \frac{2L}{D_{L,boost}^2 T_s}. \quad (6)$$

$$R_{in,SB} = \frac{v_{in}}{0.5 i_{pk} (D_{L,boost} + D_{diode})} = \frac{2L_{boost}}{D_{L,boost}^2 T_s} \left(1 - \frac{v_{in}}{V_o} \right). \quad (7)$$

Comparing (6) and (7), it is noted that R_{in} is solely a function of constant parameters. However, input resistance of

the synchronous boost rectifier $R_{in,SB}$ is not only a function of the source inductor L_{boost} , duty cycle $D_{L,boost}$, and sampling time T_s , but also depends on the voltage step-up ratio v_{in}/V_o . Therefore, if V_o is in the vicinity of v_{in} , we cannot assume $R_{in,boost} \approx 2L(D_{L,boost}^2 T_s)^{-1}$. Therefore, control methods such as input resistance control as mentioned in [11] can be implemented to compensate for the term v_{in}/V_o for synthesizing desired input resistance. The corresponding duty cycle for synchronous boost converter is the following:

$$D_{L,boost} = \left[\left((R_d + w) \left(1 - \frac{v_{in}}{V_o} \right)^{-1} \right)^{0.5} 2L_{boost} T_s \right] T_s^{-1} \quad (8)$$

where w is the output of PI controller. As indicated, measurements of both v_{in} and V_o are required in order to obtain $D_{L,boost}$. Measurement of v_{in} requires additional wide CMRR differential op-amp circuitries, which contributes to power efficiency degradation and a higher component count. On the contrary, the proposed topology does not depend on neither v_{in} nor V_o , as indicated in (6). Therefore, measurement of both variables is not necessary, therefore, no additional circuitry and power dissipation takes place.

In addition to (6), the envelope input resistance is also defined as the input voltage over the input current envelope. Since the converter is operating in DCM, after $D_{boost} T_s$, the peak value of inductor current is given by $i_{pk} = v_{in} D_{L,boost} T_s L^{-1}$. Therefore, the envelope input resistance is given by

$$R_{in_{ev}} = \frac{L}{D_{L,boost} T_s}. \quad (9)$$

From (9), it is noted that $R_{in} = 2D_{L,boost}^{-1} R_{in_{ev}}$, which is larger than the envelope input resistance by more than a factor of two. This is expected since the converter operates in DCM.

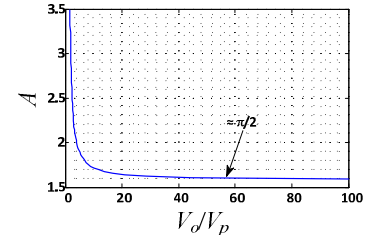


Fig. 6: Approximation of input power instantaneous term with respect to step-up ratio.

A. Line Voltage, Line Current, and Filtered Line Current

Simulation studies to verify the analytical results and performance of the proposed bridgeless converter were performed using MATLAB SIMULINK/ SimPowerSystem toolbox. The parameters, for DCM operation, were chosen as follows: fixed inductor $L = 1$ mH, inductor $r_{ESR} = 370$ mΩ, vibrational frequency $f_i = 10$ Hz, switching frequency $f_{sw} = 1$ kHz $\gg f_i$, line voltage amplitude $V_{in} = 5$ V, and battery voltage $V_o = 15$ V. Substituting into (5), $D_{L,max} = 0.6$ is obtained for ensuring DCM operation. Moreover, the physical components,

shown in Table 1, were selected for switching waveform and power efficiency simulations.

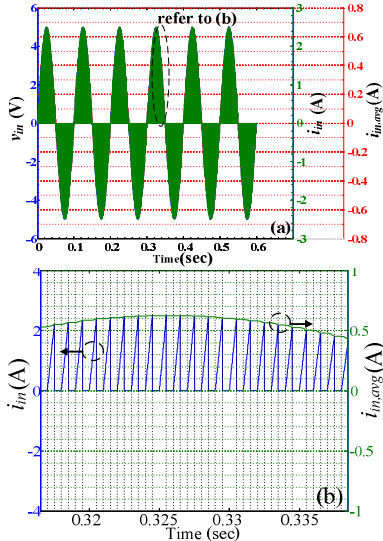


Fig. 7: (a) Simulated switching waveform of line voltage v_{in} (left x-axis), line current i_{in} (right inner x-axis), and average line current $i_{in,avg}$ (right outer x-axis), (b) Detailed switching waveform of i_{in} (left x-axis) and $i_{in,avg}$ (right x-axis) during positive cycle with duty cycle $D_L = 50\%$.

Substituting values of L , D_L , and f_{sw} into (6) and (9), the average line resistance and envelope resistance are obtained as $R_{in} = 8 \Omega$ and $R_{in,env} = 2 \Omega$, respectively. The values can be verified by taking the amplitude ratios between v_{in} , i_{in} and $i_{in,avg}$ as indicated in Fig. 7. Note $i_{in,avg}$ is obtained by averaging i_{in} every switching interval.

B. Line Voltage, Inductor and Rectifying Diode Current

According to the equivalent circuit in both Modes 1/4 and 2/3, the inductor L acts as a current source by charging from line and discharging to load. Incorporating various conduction losses during charging period (i.e. Modes 1/4) and discharging period (i.e. Mode 2/3), the input and output voltages with respect to i_{in} and i_L are approximated as the following:

$$v_{in} \approx L \frac{di_{in}}{dt} + i_{in} (r_{DS,on} + r_{ESR}) \quad (10)$$

$$V_o \approx 2 \left(L \frac{di_L}{dt} + i_L r_{ESR} + V_F \right) \quad (11)$$

where $r_{DS,on} = 0.7 \Omega$, $r_{ESR} = 0.37 \Omega$, $V_F = 0.39$ V are MOSFET on-time resistance, inductor equivalent series resistance, and diode forward voltage drop. Substituting the adopted parameters one would obtain $i_{pk} \approx 2.5$ A with $D_L = 50\%$, $f_{sw} = 1$ kHz, $L = 1$ mH, $V_{in} = 5$ V, which matches with the simulated waveforms in Fig. 8 (a). To verify i_L in Fig. 3 the detailed switching waveforms from Mode 1 to 4 in both positive and negative line cycles are shown in Fig. 8 (b) and (c).

Subsequently, the load current flowing through rectifying diodes Q_p , load capacitors C_1 , C_2 , and battery are simulated as

shown in Fig. 9. As illustrated, during positive line cycle, $i_{Qp} = |i_{C1}| + |i_{C2}|$, $i_{C1} + i_{C2} = 0$ A, and the battery current is given by $i_o = i_{C2}$. The voltage across C_1 and C_2 , shown in Fig. 10, are charged/ discharged alternatively with respect to line cycle polarities. The voltage across C_1 (i.e. v_{C1}) and C_2 (i.e. v_{C2}) sums up to V_o , which is due to opposite current flow during both discharging periods taking place in Mode- 2 and 4 as shown in Fig. 4 and Fig. 5, respectively.

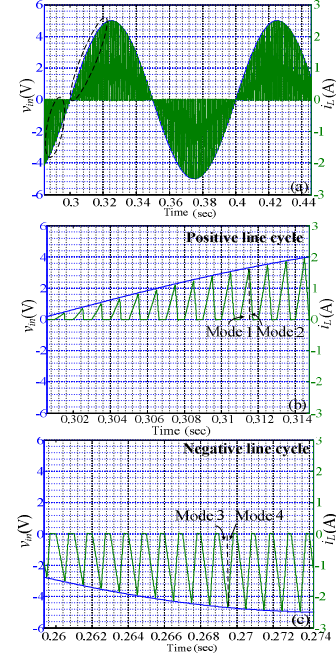


Fig. 8: (a) Simulated switching waveform of line voltage v_{in} , inductor current i_L . (b) Detailed switching waveform during positive and (c) negative line cycle.

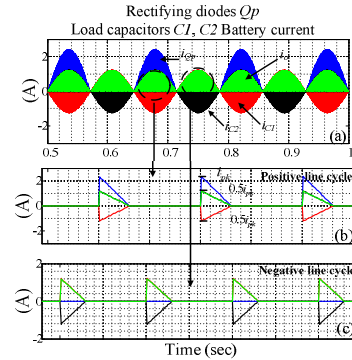


Fig. 9: (a) Simulated switching current waveform of Q_p , C_1 , C_2 , and battery current i_o . (b) Detailed switching waveform during positive and (c) negative line cycle.

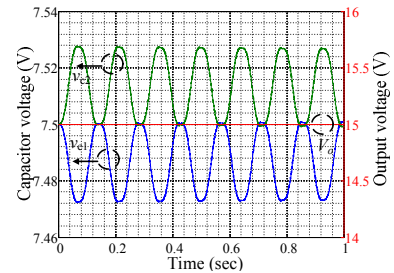


Fig. 10: Capacitors voltage ripples.

C. Power Efficiency

Converter power efficiency can be calculated by measuring converter's input and output power. Detailed estimation of power efficiency is performed by calculating the power losses of main converter components. The conduction/switching power loss of semiconductors and other passive elements are estimated using *PSIM* as depicted in Fig. 11. The thermal module allows including MOSFETs main parameters under various testing conditions such as maximum drain-source voltage V_{DS_MAX} , maximum drain current I_{D_MAX} , and switching frequency. Pre-dominant parameters such as maximum reverse recovery voltage V_{RR_MAX} , maximum continuous forward current I_{F_MAX} , and forward voltage V_F can be included for diode switching and conduction loss calculations. The power loss of passive elements can be approximated by ESR. In addition, more detailed estimation can be performed by measuring the junction thermal resistance $R_{th(j-c)}$ and $R_{th(c-s)}$, heat sink thermal resistance, and ambient temperatures.

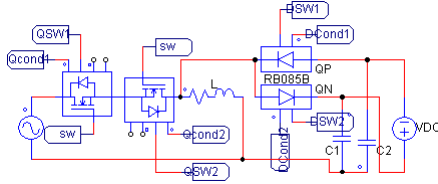


Fig. 11: Component power loss simulation in *PSIM* using thermal module.

The power efficiency is defined as the following:

$$\eta = \frac{P_{out}}{P_{in}} = 1 - \frac{P_{loss}}{P_{in}} \quad (12)$$

where P_{out} , P_{in} , and P_{loss} are the output power, input power, and power losses contributed by physical components of the power converter.

The simulated power efficiency, shown in Fig. 12, is obtained by both methods indicated in (12). The primary contributor of total power loss comes from rectifying diode conduction loss [12]. The results, obtained in *PSIM*, are plotted with respect to synthesized resistance. As depicted, both methods demonstrate highly correlated results. The highest conversion efficiency of $\approx 93\%$ at $R_{in} \approx 22 \Omega$ is obtained in simulation. In addition, the experimental power efficiencies (shown in dashed line) with $R_{in} = 10, 30, 50 \Omega$ are arbitrarily selected for experimental result presentation. The details are outlined in the following section.

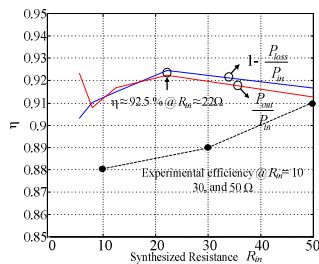


Fig. 12: Simulated and experimental power efficiency with respect to various synthesized resistances. Note: The load connects to seal lead-acid battery rated at 12 V (per battery).

Table 1. Power components selected for switching waveform and power efficiency simulation.

Component	Manufacture	Part number
n- MOSFET	Fairchild Semiconductor	FDC 8602
Power Inductor	Coilcraft	PCV- 2- 105- 02L
Electrolytic Capacitor	Nichicon	UVR1J682MRD
Schottky Diode	Fairchild Semiconductor	MBR0520
Seal Lead- acid Battery	Infinity Battery	I232/ IT 1.3- 12

V. REGENERATIVE AUTOMOTIVE SUSPENSION EXPERIMENTAL RESULTS

In order to verify resistance synthesis by the proposed converter the prototype is coupled to a 1 degree- of- freedom regenerative suspension, fabricated by Intelligent Vehicles Lab at Simon Fraser University as seen in Fig. 13. The electromagnetic- based suspension generates available AC power when placed under base excitation.

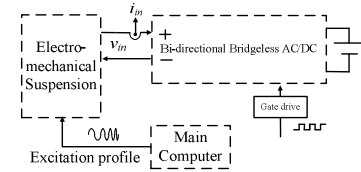


Fig. 13: Regenerative Suspension/ bi- directional bridgeless AC/DC setup.

In this work, the suspension is excited by a MTS- 248 hydraulic actuator at nominal fixed vibrational frequency and amplitude of 6 Hz and 5mm, respectively. The synthesized R_{in} is swept from 10 to 50 Ω with the increments of 20 Ω . Referring to Fig. 14 (a) both v_{in} (left y- axis) and $i_{in,avg}$ (right y- axis), by exciting the regenerative suspension for 15 (sec), are documented for synthesizing $R_{in} = 10, 30, 50 \Omega$ with $D_L \approx 45\%$, 26%, 20%, respectively. Both v_{in} and $i_{in,avg}$ are filtered by a 2nd- order low pass filter with cutoff frequency of 500 Hz.

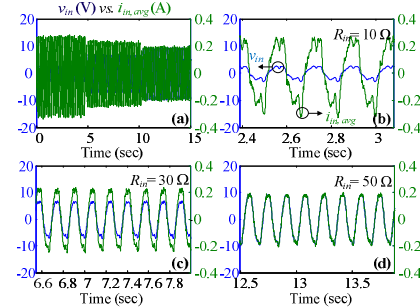


Fig. 14: (a) Sweeping synthesized R_{in} with detailed instantaneous waveform of v_{in} vs. $i_{in,avg}$ indicating (b) $R_{in} = 10 \Omega$, (c) 30Ω , and (d) 50Ω .

As depicted in Fig. 14 (b), (c) and (d) average input current $i_{in,avg}$ is observed to be in- phase (i.e. high power factor) with v_{in} while maintaining the corresponding amplitude ratio. Meaning, the converter is providing desired electrical damping by harvesting real power from the regenerative vehicle suspension. In this experiment, the average power harvested

over entire duration is roughly 0.73 (W). The available generated power through suspension vibration is roughly 0.85 (W). While regenerating from suspension prototype with various values of R_{in} the converter's conversion efficiencies η are also obtained. The experimental efficiencies for $R_{in} = 10, 30, 50 \Omega$ are $\eta \approx 87.5 \%$, 89.3% , and 91.2% , respectively.

VI. CONCLUSIONS

In this work, a new bridgeless switch-mode bidirectional converter operating in DCM is proposed for applications in low- power/ line voltage type energy harvesting. Detailed analysis and modeling of the power converter in various switching modes were presented along with control strategy for input resistance synthesis. The new topology offers simple gating pattern and lower power conduction loss compared to a conventional synchronous boost converter. According to the simulation results, the switching waveforms, such as line voltage, current, and rectified current correlated with the theoretical analysis. In addition, the simulated power efficiency of the converter operating over an extended range of synthesized resistances is reported. The ability of regulating converter input resistance can be utilized to emulate equivalent mechanical damping with regenerative capability. Experimental results of the converter connected to a regenerative vehicle suspension prototype for providing desired equivalent damping through vibrational energy regeneration is also demonstrated. Optimization of controlling vehicle dynamics and maximizing energy harvesting is the subject of future work [27].

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